MATH 211 Analysis III — Problem Session 7

Problem 1

Let

$$f(x,y) = \begin{cases} \dfrac{3xy^2}{x^2+2y^4} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0). \end{cases}$$

(a) Where is f continuous?

(b) Find the tangent plane to $z = f(x,y)$ at $(2,-1)$.

Solution. (a) For $(x,y) \neq (0,0)$, f is a quotient of polynomials and the denominator x^2+2y^4 is positive, so f is continuous at every $(x,y) \neq (0,0)$.

At the origin we approach along the path $x = y^2$:

$$f(y^2,y) = \frac{3y^2 \cdot y^2}{y^4+2y^4} = \frac{3y^4}{3y^4} = 1 \qquad (y \neq 0),$$

which does not tend to $f(0,0) = 0$. So f is continuous exactly on $\mathbb{R}^2 \setminus \{(0,0)\}$.

(b) First,

$$f(2,-1) = \frac{3 \cdot 2 \cdot 1}{4+2} = 1.$$

By the quotient rule

$$f_x = \frac{3y^2(x^2+2y^4) - 3xy^2 \cdot 2x}{(x^2+2y^4)^2} = \frac{3y^2(2y^4-x^2)}{(x^2+2y^4)^2},$$

$$f_y = \frac{6xy(x^2+2y^4) - 3xy^2 \cdot 8y^3}{(x^2+2y^4)^2} = \frac{6xy(x^2-2y^4)}{(x^2+2y^4)^2}.$$

At $(2,-1)$ we have $x^2+2y^4 = 6$, so

$$f_x(2,-1) = \frac{3 \cdot 1 \cdot (2-4)}{36} = -\frac{1}{6}, \qquad f_y(2,-1) = \frac{6 \cdot 2 \cdot (-1)(4-2)}{36} = -\frac{2}{3}.$$

Since f_x, f_y are continuous near $(2,-1)$ (rational functions with nonvanishing denominator), f is differentiable there and the tangent plane is

$$z = f(2,-1) + f_x(2,-1)(x-2) + f_y(2,-1)(y+1) = 1 - \frac{1}{6}(x-2) - \frac{2}{3}(y+1),$$

that is,

$$x + 4y + 6z = 4.$$

Problem 2

Find the tangent plane to $z = f(x,y)$ at c.

(a) $f(x,y) = x^3 \sin y$, $c = (0,0,0)$.

(b) $f(x,y) = x^3y - xy^3$, $c = (1,1,0)$.

Solution. Both functions are differentiable everywhere, and the tangent plane at (a, b) is

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

(a) $f_x = 3x^2 \sin y$ and $f_y = x^3 \cos y$. At $(0, 0)$ we get $f_x = 0$, $f_y = 0$. So the plane is

$$z = 0.$$

(b) $f(1, 1) = 0$. $f_x = 3x^2y - y^3$ gives $f_x(1, 1) = 2$, and $f_y = x^3 - 3xy^2$ gives $f_y(1, 1) = -2$. So

$$z = 2(x - 1) - 2(y - 1) \quad \Longrightarrow \quad z = 2x - 2y.$$

Alternative solution. Let $F(x, y, z) = f(x, y) - z$. The surface $z = f(x, y)$ is the level set $F = 0$, so $n = \nabla F = (f_x, f_y, -1)$ is a normal vector of the tangent plane at c. The plane through c with normal n is

$$n \cdot (x, y, z) = n \cdot c.$$

(a) $\nabla F = (3x^2 \sin y,\ x^3 \cos y,\ -1)$, so $n = \nabla F(0, 0, 0) = (0, 0, -1)$ and $n \cdot c = 0$. Hence

$$(0, 0, -1) \cdot (x, y, z) = 0 \quad \Longrightarrow \quad z = 0.$$

(b) $\nabla F = (3x^2y - y^3,\ x^3 - 3xy^2,\ -1)$, so $n = \nabla F(1, 1, 0) = (2, -2, -1)$ and $n \cdot c = 2 - 2 + 0 = 0$. Hence

$$(2, -2, -1) \cdot (x, y, z) = 0 \quad \Longrightarrow \quad 2x - 2y - z = 0 \quad \Longrightarrow \quad z = 2x - 2y.$$

Problem 3

(a) Approximate $\sqrt{(3.02)^2 + (3.97)^2}$ using a tangent plane.

(b) Approximate $(2.02)(2.97)^2\sqrt{4.04}$ in the same way, with three variables.

Solution. The idea is to pick a nearby point (a, b) where the value is easy to compute, write the tangent plane at (a, b), and plug the new (hard) point into this plane.

(a) Let $f(x, y) = \sqrt{x^2 + y^2}$. The numbers 3.02, 3.97 are close to 3, 4 and $f(3, 4) = \sqrt{9 + 16} = 5$, so the base point is $(3, 4)$.

$$f_x = \frac{x}{\sqrt{x^2 + y^2}} \ \Rightarrow\ f_x(3, 4) = \frac{3}{5}, \qquad f_y = \frac{y}{\sqrt{x^2 + y^2}} \ \Rightarrow\ f_y(3, 4) = \frac{4}{5}.$$

The tangent plane at $(3, 4)$ is

$$z = 5 + \frac{3}{5}(x - 3) + \frac{4}{5}(y - 4).$$

Plugging in $(x, y) = (3.02, 3.97)$:

$$z = 5 + \frac{3}{5}(0.02) + \frac{4}{5}(-0.03) = 5 + 0.012 - 0.024 = 4.988.$$

(b) Let $F(x, y, z) = xy^2\sqrt{z}$. The numbers 2.02, 2.97, 4.04 are close to 2, 3, 4 and $F(2, 3, 4) = 2 \cdot 9 \cdot 2 = 36$, so the base point is $(2, 3, 4)$.

$$F_x = y^2\sqrt{z} = 9 \cdot 2 = 18, \qquad F_y = 2xy\sqrt{z} = 2 \cdot 2 \cdot 3 \cdot 2 = 24, \qquad F_z = \frac{xy^2}{2\sqrt{z}} = \frac{2 \cdot 9}{2 \cdot 2} = \frac{9}{2}.$$

The tangent plane at $(2, 3, 4)$ is

$$w = 36 + 18(x - 2) + 24(y - 3) + \frac{9}{2}(z - 4).$$

Plugging in $(x, y, z) = (2.02, 2.97, 4.04)$:

$$w = 36 + 18(0.02) + 24(-0.03) + \frac{9}{2}(0.04) = 36 + 0.36 - 0.72 + 0.18 = 35.82.$$

Problem 4

Is the plane $z = 2x + 4y - 5$ tangent to the paraboloid $z = x^2 + y^2$? If so, at which point?

Solution.

Let $f(x, y) = x^2 + y^2$. $f_x(a, b)$ and $f_y(a, b)$:

$$f_x = 2a = 2 \;\Rightarrow\; a = 1, \qquad f_y = 2b = 4 \;\Rightarrow\; b = 2.$$

Then $f(1, 2) = 5$, and the tangent plane at $(1, 2)$ is

$$z = 5 + 2(x - 1) + 4(y - 2) = 2x + 4y - 5,$$

which is exactly the given plane. So yes, it is tangent, at the point $(1, 2, 5)$.